

Argus Demo Script — Under 10 Minutes

A concise walkthrough for presenting **Argus** (Adaptive Stigmergic Oversight for AI Agent Swarms) to students. Based on the [ASO paper](#).

Before the Demo (Setup)

1. Prerequisites

- Node.js 18+
- `argus.config.local.yaml` with `repository` set to a GitHub repo you can push to
- `CURSOR_API_KEY` env var (or `apiKeyPath` in config)
- Optional: `GITHUB_TOKEN` for cleanup

2. Build

```
npm install && npm run build
```

3. Optional: Clean slate (if re-running)

```
export GITHUB_TOKEN=ghp_...  
npx argus cleanup
```

Demo Flow (~10 min)

1. Introduction (30 sec)

"Argus lets one human operator supervise many AI coding agents. It uses three ideas from the ASO paper: **intent delegation** (you specify what, not how), **stigmergic coordination** (agents work via a shared Git repo), and **exception-based review** (you only intervene when something needs attention)."

2. Launch a Swarm (1 min)

```
npx argus run intents/oauth2-auth.intent.yaml
```

- Explain: "This decomposes the intent into 5 work packages and launches 5 Cursor agents in parallel."
 - The dashboard opens automatically at **`http://localhost:3848`** — open it in a browser.
 - Keep the terminal visible so students see logs.
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3. Intent Delegation Tab (2–3 min)

1. Left panel — Job list

- Point out the new job with status RUNNING.
- Show intent summary and "5 agents".

2. Right panel — Pipeline

- Walk through stages:
 - **Intent Defined** — human specifies the goal
 - **Decomposer** — breaks it into work packages (Explorer, Workers, Validator)
 - **Orchestrator** — agents run in parallel (spinning = running)
 - **Calculating Stigmergic Metrics** — fan-out, exception rate, trust
 - **Create Exception Review** — validation and human review queue

3. New Job

- Click "+ New Job".
 - Show "From File" vs "Inline" modes.
 - Show trust thresholds (auto-approve, escalate, block).
 - Cancel or launch a second job if time allows.
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4. Swarm & Exception Review Tab (2–3 min)

1. Switch to Swarm tab (icon in sidebar).

2. Metrics strip

- **Fan-out** — effective swarm size
- **Exception rate** — % needing human review
- **Throughput/hr** — agents finished per hour
- **Trust mean** — average agent trust score
- **Containment** — % auto-approved vs total

3. Agent Swarm grid

- Dots = agents (green = finished, blue = running, red = error, orange = blocked).
- Hover a dot to see work package details and PR link.

4. Exception Review Queue

- Exceptions = low-confidence or blocked agents.
- Use "+ Test" to add a synthetic exception for demo.
- Approve / Reject / Follow-up for blocked agents.

5. Activity feed

- Recent agent completions with timestamps.
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5. Key Concepts (1 min)

Concept	One-liner
Intent delegation	Human defines <i>what</i> ; decomposer and agents handle <i>how</i>
Stigmergic coordination	Agents coordinate via shared Git (branches, PRs)
Exception-based review	Human intervenes only when validation flags issues
Trust scores (τ)	Per-agent confidence; low $\tau \rightarrow$ more review

6. Wrap-up & Cleanup (1 min)

- **Stop the run:** Ctrl+C in the terminal.
- **Inspect later:** `npx argus ui` $\rightarrow$ `http://localhost:3848`
- **Re-run from clean repo:**

```
npx argus cleanup
npx argus run intents/oauth2-auth.intent.yaml
```

Quick Reference

Action	Command / URL
Run swarm + UI	<code>npx argus run intents/oauth2-auth.intent.yaml</code>
UI only	<code>npx argus ui</code> $\rightarrow$ <code>http://localhost:3848</code>
List intents	<code>npx argus intent list</code>
Clean branches	<code>npx argus cleanup</code> (needs <code>GITHUB_TOKEN</code>)
Exception test intent	<code>intents/exception-test.intent.yaml</code>

Troubleshooting

- **Port in use:** `fuser -k 3848/tcp` then restart.
- **No exceptions:** Use "+ Test" in Exception Review Queue, or run `exception-test-strict.intent.yaml`.
- **API key:** Ensure `CURSOR_API_KEY` or `apiKeyPath` is set; see [README](#).